

## USER QUERY & CANDIDATE APIS

**[Kiểm tra phạt nguội cho xe máy biển số 59P1-12345 tại TP. Hồ Chí Minh]**

[EN: Check traffic fines for motorbike 59P1-12345 in HCMC]

**Candidate Tools:** [tra\_cuu\_phat\_nguoi, dat\_ve\_xe\_khach, thanh\_toan\_tien\_dien,...]

### Method 1: End-to-End Generative SLM

Qwen3.5 (2B/4B) + QLoRA SFT (Response-only loss)

Autoregressive Decoding (Native XML tags)

Direct JSON Tool Call Output

### Method 2: Bi-Encoder + Cross-Encoder

Dense Retrieval (BGE-M3 + CachedMNRL) → Top-3 Candidates

Abstention & Threshold Gate ( $\tau=0.35$ ,  $\delta=0.21$ )

Hierarchical Cross-Encoder (XLM-R)

 has\_value

 Span Extraction

 Enum/Bool

Value Normalizer & Schema Validation → Deterministic JSON